

REAL ESTATE ANALYTICS AND RECOMMENDATION SYSTEM
MAJOR PROJECT REPORT

**Submitted In Partial Fulfilment Of The Requirements For
The Award Of The Degree Of
Bachelor Of Technology Artificial Intelligence And Machine
Learning**



**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA,
(State Technological University of M.P.)**

Submitted By:

Mukul Bandewar	(0133CL221080)
Pawan Viswakarma	(0133CL221094)
Raghupat Pal	(0133CL221099)
Ramchand Sevaiwar	(0133CL221106)

Under The Supervision of

Prof. Dr. Rupali

DEPARTMENT OF AIML



DEPARTMENT OF AIML

SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL(M.P.)



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL
Department of AIML

DECLARATION

I hereby declare that the work, which is being presented in the major project, entitled **“REAL ESTATE ANALYTICS AND RECOMMENDATION SYSTEM”** submitted in the **Department of AIML, Sagar Institute of Research & Technology, Bhopal** is an authentic record of my own work carried out during the period from July 2025 – June 2026, under the Guidance of Asst. **Prof. Dr. Rupali** Department of AIML, Sagar Institute of Research & Technology, Bhopal.

Date:

Place: Bhopal

Mukul Bandewar (0133CL221080)

Pawan Vishwakarma (0133CL221094)

Raghupat Pal (0133CL221099)

Ramchand Sevaiwar (0133CL221106)



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL
Department of AIML

CERTIFICATE

This is to certify that the Major Project “**REAL ESTATE ANALYTICS AND RECOMMENDATION SYSTEM**” being Submitted by **Mukul Bandewar (0133CL221080) Pawan Vishwakarma (0133CL221094) Raghupat Pal (0133CL221099) Ramchand Sevaiwar (0133CL221106)** in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology** in AIML to **RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL(M.P.)** during the academic year **2025-26** is a record of bona fide piece of work, carried out by her under my supervision and guidance in the **Department of AIML, Sagar Institute of Research & Technology, Bhopal.**

Guide Name

Prof. Dr. Rupali

Project Coordinator

Prof. Menali Paul

HoD AIML

Dr. Kalpana Rai



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL
Department of AIML

APPROVAL CERTIFICATE

The major project entitled **“REAL ESTATE ANALYTICS AND RECOMMENDATION SYSTEM”** being submitted by **Mukul Bandewar (0133CL221080) Pawan Vishwakarma (0133CL221094) Raghupat Pal (0133CL221099) Ramchand Sevaiwar (0133CL221106)** to **RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL(M.P.)** has-been examined by us and is hereby approved for the award of degree **“Bachelor of Technology in AIML”** in the academic year **2025-26**.

(Internal Examiner)

Date: _____

(External Examiner)

Date: _____



SAGAR INSTITUTE OF RESEARCH & TECHNOLOGY, BHOPAL

Department of AIML

ACKNOWLEDGEMENT

This is one of the best moments of my B.Tech. program to publicly acknowledgment those who have contributed in many ways to make my success apart of their own. The completion of the Major Project depends upon the co-operation, coordination and combined effects of several resources of knowledge energy.

I heartily thanks to **Prof. Dr.Rupali** and **Prof. Menali Paul**, who have supported in Major project, faculties of Department of AIML, for accepting me to work under their Valuable Guidance, Closely Supervised this work over the past few months and offering many innovative ideas and helpful suggestions, which led to the successful completion of this dissertation work.

I am especially thankful to **Dr. Rajiv Srivastava**, Director (SIRT, Bhopal) for his kind co-operation and rendering all possible facilities.

I express my thanks to **Dr. Kalpana Rai**, HOD AIML department SIRT Bhopal for kind support.

I am thankful to all staff members of the AIML department and my friends for their timely help, co-operation and suggestions during my project work. Lastly but not the last, I must express thanks to my family, without their moral support it was impossible for me to complete this major project work.

Mukul Bandewar (0133CL221080)

Pawan Vishwakarma (0133CL221094)

Raghupat Pal (0133CL221099)

Ramchand Sevaiwar (0133CL221106)

Abstract

The Real Estate Analytics and Recommendation System is a smart web-based application developed to simplify property searching, analysis, and investment decision-making. With the rapid growth of the real estate sector, finding suitable properties based on user requirements has become a challenging task due to the large amount of available data. This project aims to provide an intelligent platform that analyzes real estate data and recommends the most suitable properties to users based on their preferences and budget.

The system collects and processes property-related information such as location, price, property type, amenities, area, and market trends. By utilizing data analytics and machine learning algorithms, the system identifies patterns in property prices and predicts future market behavior. The recommendation engine generates personalized property suggestions by considering user preferences, search history, and investment goals. Additionally, the system provides comparative analysis, price trend visualization, and location-based insights to assist users in making informed decisions.

The proposed system benefits property buyers, sellers, investors, and real estate agencies by reducing manual effort, improving search accuracy, and enhancing transparency in the property market. The integration of predictive analytics and recommendation techniques enables users to discover profitable investment opportunities and suitable residential or commercial properties efficiently. This project demonstrates how modern data analytics and intelligent recommendation systems can transform the traditional real estate decision-making process into a faster, more reliable, and user-friendly experience.

The system enables users to search and compare properties based on factors such as budget, location, property type, infrastructure facilities, and investment potential. Predictive analytics are employed to estimate future property values and market trends, helping investors make profitable decisions. The recommendation engine suggests properties that best match user requirements by analyzing historical user behavior and preference data.

Additionally, the platform provides interactive dashboards and visualizations to present market insights, price fluctuations, and demand analysis in an easy-to-understand format. By automating property analysis and recommendation processes, the system improves decision-making efficiency, reduces search time, and enhances user satisfaction.

Keywords: Real Estate Analytics, Recommendation System, Machine Learning, Property Prediction, Data Analytics, Market Trends, Personalized Recommendations, Real Estate Investment.

List of Figures

Fig 3.1 System Architecture layer.....	13
Fig 3.2 Use Case Diagram.....	16
Fig.3.3 Class Diagram.....	17
Fig.3.4 Activity Diagram.....	18
Fig.3.5 Sequence Diagram.....	19
Fig.3.6 Flow Chart Diagram.....	20
Fig.4.3 Project Scheduling (Gantt Chart).....	25
Fig.4.4 PERT Chart.....	26
Fig.4.5 Workflow 1.....	27
Fig.4.6 Workflow2.....	28

List of Tables

Table no. 4.1 Development tool.....	23
Table no.4.5 Test Cases.....	32
Table no. 5.2.Search Input.....	36
Table no. 5.4.Analysis Dashboard.....	38
Table no. 5.5. System Performance.....	40

Table of Content

Contents

FRONT PAGE.....	i
DECLARATION	ii
CERTIFICATE	iii
APPROVAL CERTIFICATE	iv
ACKNOWLEDGEMENT	v
Abstract	vi-vii
List of Figures	viii
List of Tables	ix
Table of Content	x-xi
Chapter 1 Introduction.....	1
1.1 Project Category.....	2
1.2 Objectives	2
1.3 Problem Formulation	2
1.4 Identification of Need	3
1.5 Existing System	3
1.6 Proposed System	4
1.7 Unique Features of the System	4
Chapter 2 Requirement Analysis and System Specification.....	5
2.1 Software Requirement Specification (SRS).....	5
2.2 Validation.....	8
2.3 Expected Hurdles	9
2.4 SDLC Model to be Used.....	10
Chapter 3 Real State Analytics and Recommendation system	13
3.1 Architecture Overview	13
3.2 Key Components.....	14
3.3 Structured Diagrams.....	14

3.4 Dataset and Preprocessing	14
3.5 Training Process	14
3.6 Performance and metrics.....	15
3.7 Code Summary.....	15
Chapter 4 Implementation, Testing, and Maintenance	21
4.1 Languages, Tools, and Technologies	21
4.2 Coding Standards	23
4.3 Project Scheduling	25
4.4 System Implementation.....	27
4.5 Testing	30
4.6 Maintenance Strategy.....	33
Chapter 5 Results and Discussions	34
5.1 User Interface Representation	34
5.2 Property Search Module Results	35
5.3 Recommendation System Results	36
5.4 Analytics Dashboard Results	37
5.5 System Performance Analysis	39
5.6 Snapshots with Description	40
Chapter 6 Conclusion and Future Scope	43
6.1 Conclusion	43
6.2 Future Scope.....	44
Chapter 7 References / Bibliography	46
7.1 Books and Research Paper.....	48
Chapter 8 APPENDIX : A Source Code Registry	50

Chapter 1 :Introduction

The real estate sector plays a vital role in the economic development of a country by providing residential, commercial, and industrial properties to individuals and organizations. With rapid urbanization and increasing demand for properties, the real estate market has become highly competitive and data-intensive. Buyers, sellers, and investors are required to evaluate numerous factors such as property prices, location advantages, amenities, infrastructure facilities, and future market trends before making decisions. However, the availability of a large number of property listings often makes the selection process complex and time-consuming.

Traditional real estate platforms mainly focus on displaying property listings and basic search functionalities. These systems provide limited support for analyzing market trends and understanding user preferences. As a result, users often spend considerable time comparing properties manually and may not always identify the most suitable investment opportunities. The absence of intelligent decision-support mechanisms creates challenges in finding properties that best match individual requirements and financial constraints.

Advancements in data analytics, machine learning, and web technologies have created opportunities to develop intelligent systems capable of analyzing large volumes of real estate data and generating meaningful insights. By utilizing historical property records, market trends, user behavior, and location-based information, such systems can provide accurate recommendations and assist users in making informed decisions. Recommendation systems have already proven successful in domains such as e-commerce, entertainment, and online retail, and their application in real estate can significantly improve property discovery and investment planning.

The Real Estate Analytics and Recommendation System is designed to address these challenges by integrating property management, data analytics, and recommendation techniques into a single platform. The system collects and stores comprehensive property information, including location, price, property type, area, amenities, and availability status. Using analytical methods, it evaluates market trends, identifies pricing patterns, and predicts potential property values. Based on user preferences such as budget, location, and desired features, the recommendation engine suggests the most relevant properties. The proposed system not only simplifies property searching but also provides valuable market insights through visual reports, charts, and comparative analyses. It enables users to compare multiple properties, understand market behavior, and identify profitable investment opportunities. By reducing manual effort and improving recommendation accuracy..

1.1 Project Category

The Real Estate Analytics and Recommendation System falls under the category of Data Analytics and Intelligent Decision Support Systems. It combines web technologies, database management, and recommendation algorithms to provide users with meaningful property suggestions and market insights. The project focuses on analyzing real estate data and assisting users in making informed decisions regarding property purchases, sales, and investments.

1.2 Objectives

The main objectives of the project are:

1. To develop a centralized platform for managing real estate property information.
2. To provide personalized property recommendations based on user preferences.
3. To analyze property data and identify market trends.
4. To enable users to search and compare properties efficiently.
5. To assist investors in making informed decisions through analytical insights.
6. To reduce the time and effort required for property selection.
7. To improve transparency and accessibility in the real estate market.

1.3 Problem Formulation

The real estate industry involves large volumes of property listings with varying prices, locations, and amenities. Users often face difficulties in finding properties that match their requirements due to information overload and lack of proper analytical tools. Traditional property search methods rely heavily on manual filtering and broker recommendations, which may not always provide optimal results.

There is a need for an intelligent system capable of analyzing property data, understanding user preferences, and recommending the most relevant properties while also providing market insights and predictive analysis.

1.4 Identification of Need

The need for this project arises from several challenges faced by property buyers, sellers, and investors:

- Difficulty in finding suitable properties among thousands of listings.
- Lack of personalized recommendations.
- Limited access to market trend analysis.
- Time-consuming property comparison processes.
- Inadequate decision-support tools for investment planning.
- Increasing demand for digital real estate solutions.

A smart recommendation and analytics platform can significantly improve the efficiency and accuracy of property-related decisions.

1.5 Existing System

Currently, many real estate websites provide property listings and basic search functionality. Users can search properties based on location, budget, and property type. However, most existing systems have the following limitations:

- Limited personalization features.
- Absence of advanced recommendation mechanisms.
- Lack of predictive analytics for future price estimation.
- Insufficient market trend visualization.
- Dependence on manual property filtering.
- Limited comparative analysis among properties.

As a result, users often spend significant time evaluating multiple properties before making a decision.

1.6 Proposed System

The proposed Real Estate Analytics and Recommendation System aims to overcome the limitations of existing platforms by incorporating intelligent analytics and recommendation features.

The system will:

- Store and manage property data in a centralized database.
- Allow users to search properties using multiple criteria.
- Recommend properties based on user preferences and behavior.
- Analyze historical property data to identify trends.
- Generate visual reports and dashboards.
- Provide comparative analysis of selected properties.
- Support investment decision-making through predictive insights.

The platform will offer a simple and user-friendly interface accessible through web browsers.

1.7 Unique Features of the System

The proposed system includes several unique features:

Personalized Recommendations

Provides customized property suggestions based on user preferences, budget, and search history.

Market Trend Analysis

Analyzes historical data to identify trends in property prices and demand.

Predictive Property Valuation

Estimates future property values using analytical models.

Interactive Dashboard

Displays graphs, charts, and reports for easy understanding of market conditions.

Advanced Property Search

Supports filtering based on multiple criteria such as location, budget, area, amenities, and property type.

Property Compariso

Chapter 2: Requirement Analysis and System Specification

Requirement Analysis and System Specification is one of the most important phases in software development. It involves identifying user needs, defining system functionalities, and specifying both functional and non-functional requirements. Proper requirement analysis helps ensure that the developed system meets user expectations and performs efficiently in real-world scenarios.

The Real Estate Analytics and Recommendation System is designed to provide intelligent property recommendations and analytical insights based on user preferences and market data. During this phase, the requirements of buyers, sellers, administrators, and investors are carefully analyzed to determine the system architecture, database structure, user interface design, and technology stack. The specification serves as a blueprint for the development, testing, and deployment of the project.

The system must be capable of managing large amounts of property information, performing analytical operations, generating recommendations, and presenting results through an intuitive interface. It should also maintain data security, reliability, and scalability to accommodate future growth.

2.1 Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) document defines the functional and non-functional requirements of the proposed system.

Functional Requirements

User Registration and Login

- Users can create accounts and log in securely.
- Authentication and authorization mechanisms are implemented.

Property Management

- Add, update, delete, and view property listings.
- Store details such as price, location, property type, area, and amenities.

Property Search

- Search properties based on:
 - Location
 - Budget
 - Property type
 - Area size
 - Amenities

Recommendation Module

- Recommend properties according to user preferences.
- Analyze user search behavior and interests.

Analytics Dashboard

- Display market trends and property statistics.
- Provide graphical representations of property data.

Property Comparison

- Compare multiple properties simultaneously.
- Highlight differences in price, location, and features.

Administration Module

- Manage users and property listings.
- Monitor system activities and reports.

Performance

- Fast response time for search and recommendation operations.
- Efficient handling of large datasets.

Reliability

- Accurate storage and retrieval of information.
- Consistent system performance.

Security

- Secure authentication mechanisms.
- Protection against unauthorized access.
- Data encryption for sensitive information.

Scalability

- Support increasing numbers of users and properties.

Usability

- User-friendly interface.
- Easy navigation and accessibility.

Maintainability

- Modular design for future updates and enhancements.

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: 250 GB or above
- Internet Connection

Software Requirements

- Operating System: Windows/Linux
- Frontend: HTML, CSS, JavaScript
- Backend: Python (Django/Flask) or PHP
- Database: MySQL
- IDE: VS Code / PyCharm
- Browser: Chrome, Firefox, Edge

2.2 Feasibility Study

A feasibility study evaluates whether the project can be successfully implemented from technical, economic, operational, and schedule perspectives.

Technical Feasibility

The proposed system can be developed using modern web technologies and database management systems. Recommendation algorithms and analytical techniques are readily available through various machine learning libraries. Therefore, the project is technically feasible.

Economic Feasibility

The development cost is relatively low because most technologies used are open-source. Minimal hardware resources are required, making the project economically viable.

Operational Feasibility

The system provides an intuitive user interface that can be easily used by buyers, sellers, and administrators. Proper training requirements are minimal, ensuring smooth adoption.

Schedule Feasibility

The project can be completed within the allocated academic project timeline through systematic planning and development phases.

2.3 Validation

Validation ensures that the developed system satisfies user requirements and performs according to specified objectives.

The validation process includes:

Requirement Validation

- Verification of user needs and project objectives.
- Confirmation of functional requirements.

System Validation

- Ensuring all modules operate correctly.
- Verification of recommendation outputs.

User Validation

- Collecting user feedback.
- Measuring user satisfaction levels.

Data Validation

- Ensuring accuracy and consistency of stored property information.
- Preventing duplicate or invalid entries.

Performance Validation

- Testing response times under different workloads.
- Evaluating system efficiency and scalability.

2.4 Expected Hurdles

Several challenges may arise during the development and deployment of the system.

Data Availability

Obtaining large and accurate real estate datasets can be difficult. Incomplete or outdated data may affect recommendation quality.

Data Quality Issues

Missing values, duplicate records, and inconsistent formatting require extensive preprocessing before analysis.

Recommendation Accuracy

Generating highly accurate recommendations requires effective algorithms and sufficient historical data.

Market Variability

Property prices and market conditions change frequently, making prediction and analysis more complex.

Security Concerns

Protecting user data and preventing unauthorized access are important challenges that must be addressed.

System Scalability

As the number of users and property listings grows, maintaining system performance becomes increasingly important.

User Acceptance

Users may initially be hesitant to rely on automated recommendations for property decisions.

Addressing these hurdles through proper planning, testing, and optimization is essential for successful project implementation.

2.5 SDLC Model Used

The Waterfall SDLC Model is used for the development of the Real Estate Analytics and Recommendation System because it follows a structured and sequential approach suitable for academic projects.

Phases of Waterfall Model

1.Requirement Gathering and Analysis

- Identify project objectives.
- Gather user requirements.
- Prepare Software Requirement Specification (SRS).

2. System Design

- Design system architecture.
- Create database schema.
- Design user interfaces and workflows.

3. Implementation

- Develop frontend and backend modules.
- Implement recommendation and analytics features.
- Integrate database functionalities.

4. Testing

- Conduct unit testing, integration testing, and system testing.
- Verify functionality and performance.

5. Deployment

- Deploy the application in the target environment.
- Configure database and server settings.

6. Maintenance

- Fix bugs and errors.
- Update system features.
- Improve performance based on user feedback.

Advantages of Using Waterfall Model

- Simple and easy to understand.
- Well-defined stages and documentation.
- Suitable for projects with clearly defined requirements.
- Easier project management and progress tracking.

Chapter 3: Real Estate Analytics and Recommendation System

3.1 System Architecture Overview

- **Front-End / UI:** User-facing portals for buyers/sellers.
- **Back-End Application Server:** Handles user requests, fetches property details, and runs recommendation business logic.
- **Database & Big Data Layer:** Stores scraped and real-time data using scalable storage systems. [6, 7]



Fig .3.1 System Architecture layer

3.2 Modules Description

- **User Authentication & Profile:** Secure account management and user preferences tracking.
- **Search & Filter:** Allows users to find properties by location, price range, and amenities.
- **Analytics Dashboard:** Visualizes market trends, property valuation forecasts, and demographic data. [[8](#), [9](#)]

3.3 Data Collection and Database Design

- **Data Gathering:** Sources data from MLS (Multiple Listing Services), web scraping, and user interaction logs.
- **Database Schema:** Structured using three-level DBMS architecture (External, Conceptual, and Internal) to ensure data independence.

3.4 Property Data Preprocessing

- **Data Cleansing:** Handles missing values, removes duplicates, and standardizes formats (e.g., area measurements).
- **Feature Engineering:** Extracts critical variables like property age, proximity to transit, and price per square foot.

3.5 Recommendation Engine Design

- **Collaborative Filtering:** Recommends properties based on the behavioral patterns and preferences of similar users.
- **Content-Based Filtering:** Matches property features (number of bedrooms, square footage) to user search history.

3.6 Analytics and Prediction Models

- **Price Prediction:** Uses machine learning regression techniques (e.g., multiple linear regression) to estimate property market values.
- **Market Trend Analysis:** Analyzes historical sales data and current economic indicators to forecast future real estate price trends. [[2](#), [10](#)]

3.7 UML Diagrams and Structured Design

- **Use Case Diagram:** Maps out user interactions (buyers, sellers, and admins) with system functionalities.

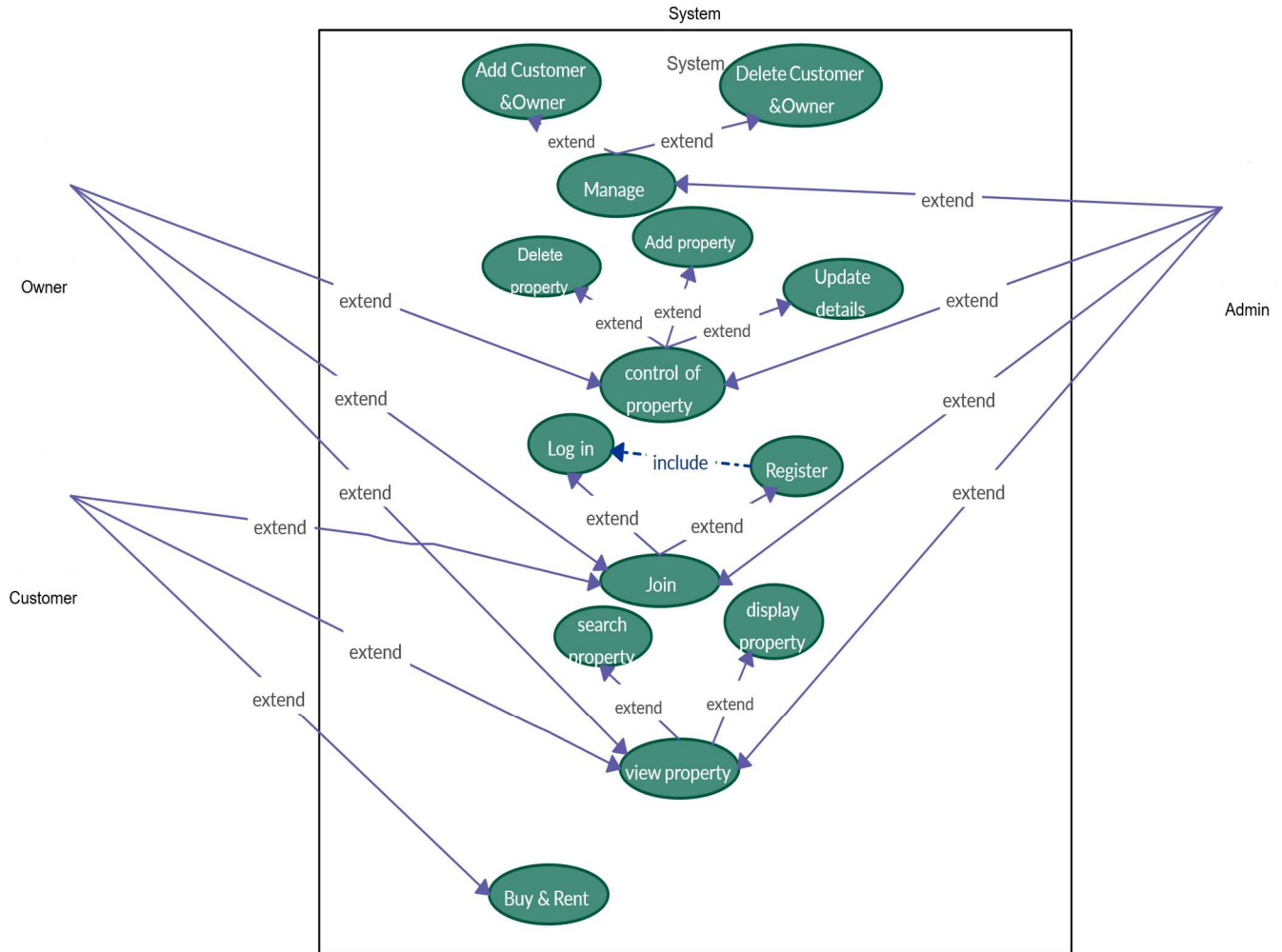


Fig 3.2 Use Case Diagram

Class Diagram: Defines the static structure, including , , , and classes.

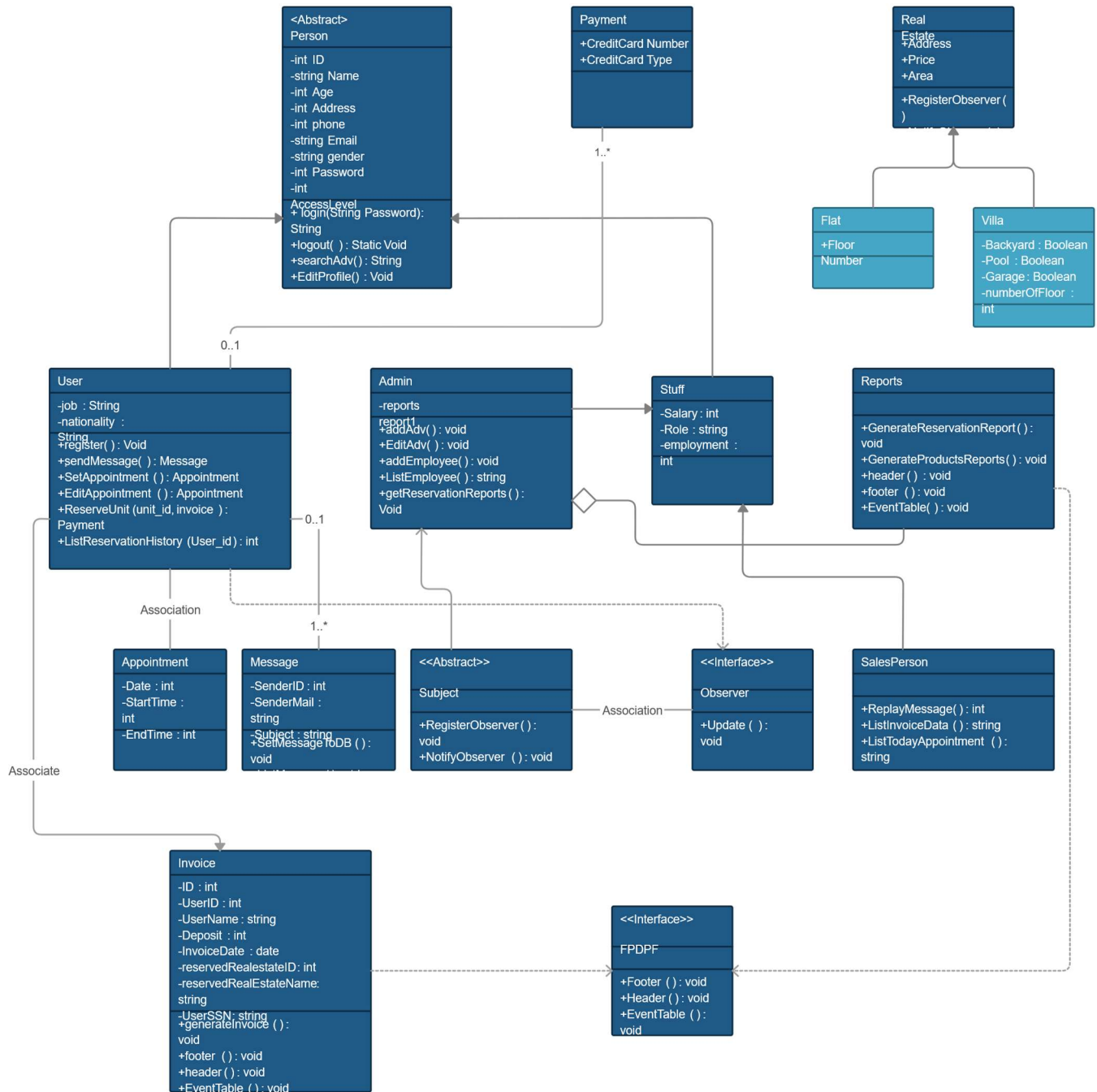


Fig.3.3 Class Diagram

- **Activity Diagram:** Illustrates the workflow of a user searching for a property and saving it as a favorite.

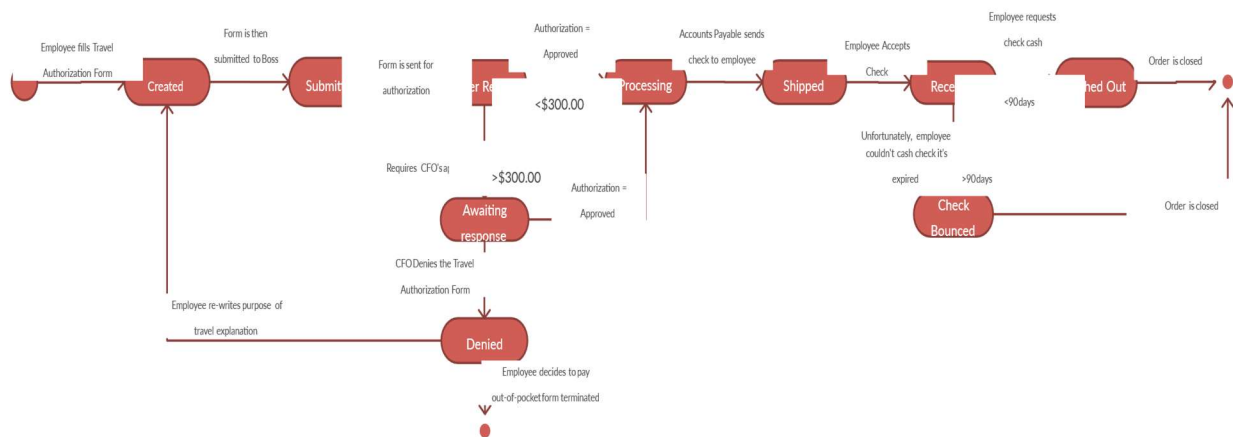


Fig.3.4 Activity Diagram

Sequence Diagram: Shows the chronological message flow between objects when a user requests a property recommendation.

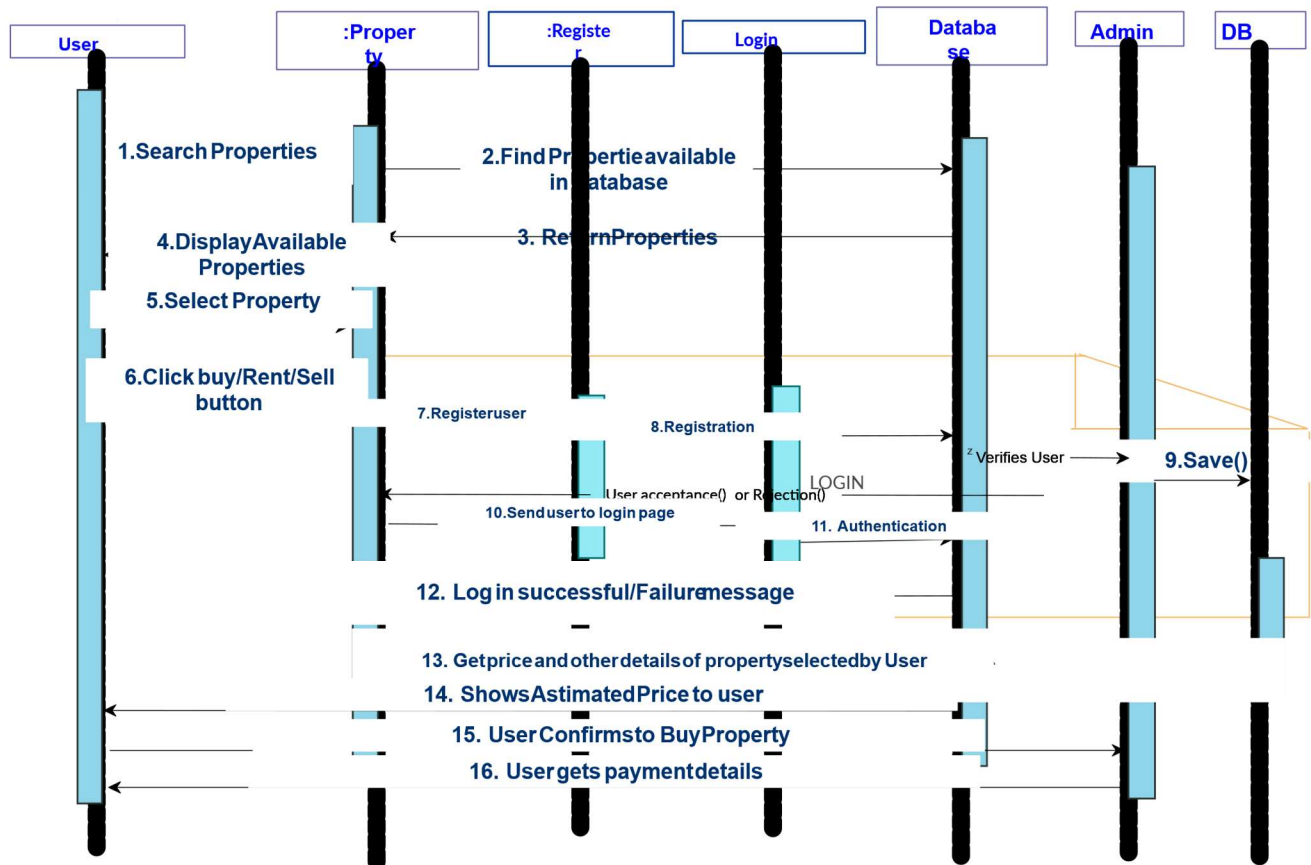


Fig.3.5 Sequence Diagram

Data Flow Diagram (DFD): Tracks how input property data flows through preprocessing to the recommendation module.

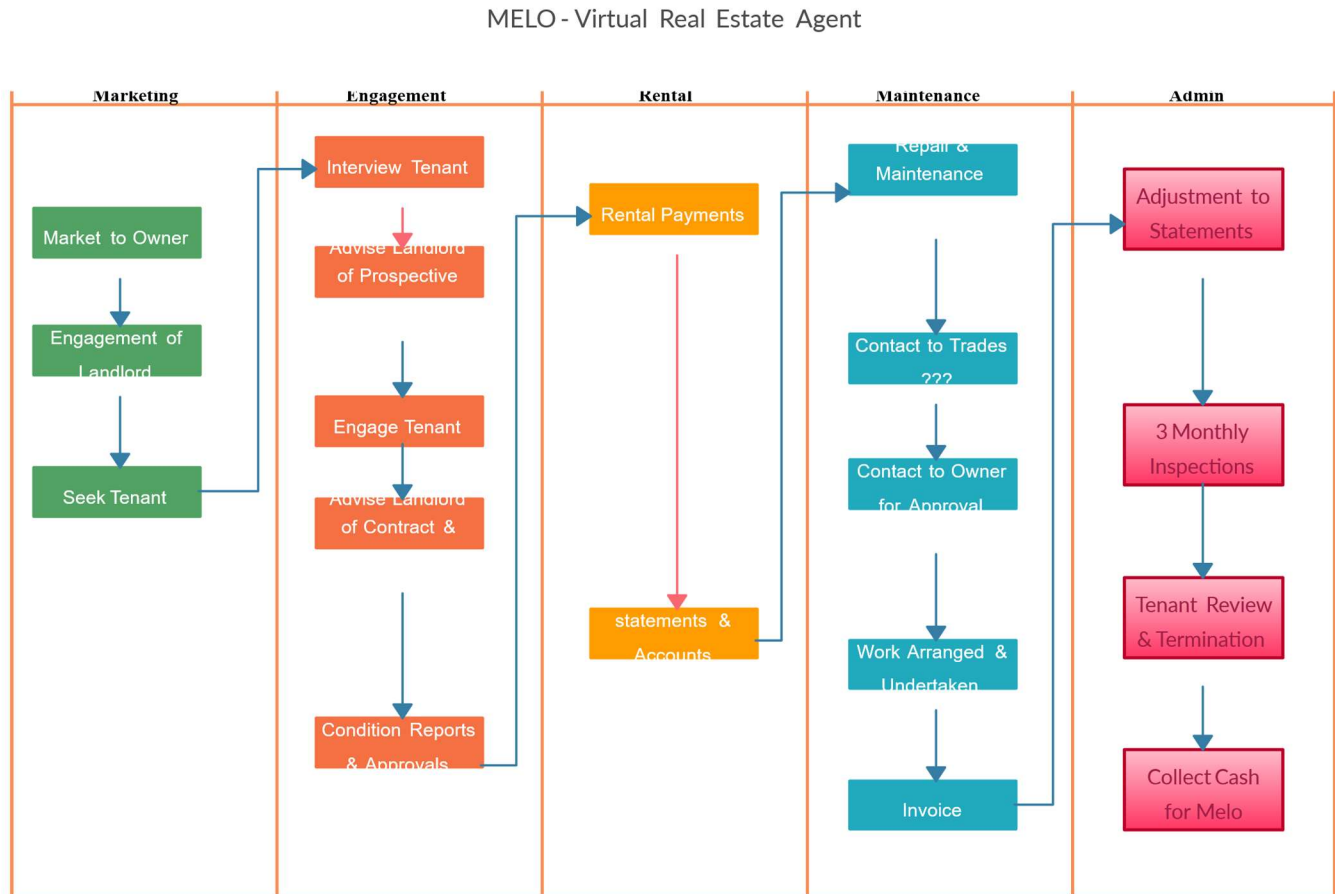


Fig.3.6 Flow Chart Diagram

Chapter 4: Implementation, Testing And Maintenance

The implementation phase transforms the system design into a fully functional software application. This phase involves coding, database creation, module integration, user interface development, and deployment. After implementation, extensive testing is performed to ensure that the system meets all functional and non-functional requirements. Maintenance activities are also planned to support future enhancements, bug fixes, and performance improvements.

The Real Estate Analytics and Recommendation System has been developed using modern web technologies and database management systems. The implementation focuses on providing a user-friendly interface, efficient data processing, secure authentication, accurate recommendations, and interactive analytics dashboards.

4.1 Languages, Tools and Technologies Used

The successful implementation of the project requires a combination of programming languages, frameworks, databases, and development tools.

Programming Languages

Python

Python is used for backend development, recommendation algorithms, data analysis, and prediction models.

HTML

Used to create the structure of web pages.

CSS

Used for designing and styling the user interface.

JavaScript

Provides interactivity and dynamic functionality within the application.

SQL

Used for database creation, querying, and management.

Frameworks and Libraries

Django / Flask

Backend web framework for handling application logic and routing.

Pandas

Used for data preprocessing and analysis.

NumPy

Provides numerical computation capabilities.

Scikit-learn

Used for recommendation and prediction models.

Matplotlib / Plotly

Used to generate graphs and visualizations for analytics dashboards.

Database

MySQL

Stores:

- User information
- Property listings
- Recommendation records
- Analytics data

Development Tools

Tool	Purpose
Visual Studio Code	Code Development

Tool	Purpose
MySQL Workbench	Database Management
GitHub	Version Control
Chrome Browser	Testing and Execution
Draw.io	UML and Flowchart Design

Table.4.1. Development tool

4.2 Coding Standards

Coding standards are followed to improve readability, maintainability, and consistency throughout the project.

Naming Conventions

Variables

Use meaningful variable names.

Example:

```
property_price = 5000000
user_location = "Bhopal"
```

Functions:

```
def search_property():
```

```
    pass
```

Function names use lowercase letters and underscores.

Classes:

```
class RecommendationEngine:
```

Class names use PascalCase.

Code Documentation

Comments are added where necessary.

Example:

```
filtered_properties = property_data[property_data['price'] <= budget]
```

Indentation and Formatting

- Four-space indentation is maintained.
- Proper spacing around operators.
- Consistent bracket placement.

Error Handling:

Example:

try:

```
user = login(username, password)
```

except Exception:

```
print("Invalid Login Credentials")
```

Proper exception handling improves system reliability.

Security Standards

- Password encryption
- Input validation
- SQL injection prevention
- Session management

These standards help maintain software quality and security.

4.3 Project Scheduling (Gantt Chart)

Project scheduling ensures timely completion of development activities.

Project Timeline

Activity	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Requirement Analysis	■	■						
System Design		■	■					
Database Design		■	■					
Coding & Development			■	■	■			
Recommendation Module				■	■			
Testing					■	■		
Documentation						■	■	
Deployment							■	
Final Review								■

Fig.4.3 Project Scheduling (Gantt Chart)

PERT Chart (Conceptual)

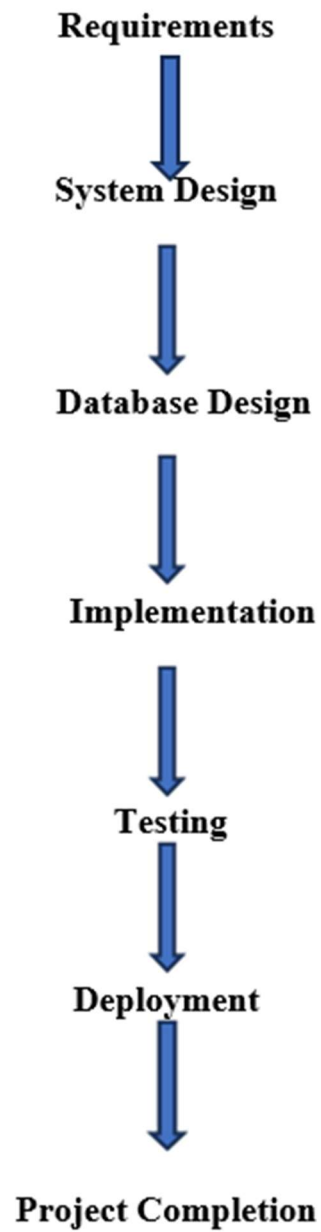


Fig.4.4 PERT Chart

4.4 System Implementation

The implementation process consists of developing and integrating multiple modules.

User Authentication Module

Features:

- Registration
- Login
- Password validation
- Session management

Workflow



Fig.4.5 Workflow 1

Property Management Module

Functions:

- Add property
- Update property
- Delete property
- View property details

Workflow

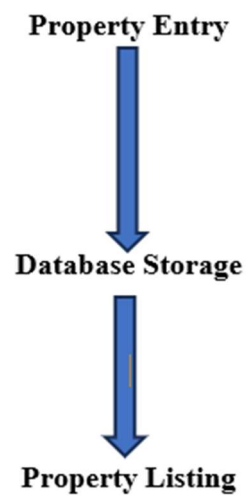


Fig4.6. Workflow 2

Search and Filter Module

Allows users to search properties using:

- Budget
- Location
- Property Type
- Amenities

Results are generated instantly from the database.

Recommendation Engine Module

Steps:

1. Collect user preferences.
2. Filter matching properties.
3. Calculate relevance score.
4. Rank properties.
5. Display recommendations.

Analytics Dashboard Module

Displays:

- Market trends
- Price distributions
- Property demand analysis
- Investment insights

Charts and graphs provide visual understanding of market behavior.

4.5 Testing Methodology

Testing ensures that the system functions correctly and meets specified requirements.

Unit Testing

Individual modules are tested independently.

Example

- Login functionality
- Search functionality
- Recommendation generation

Integration Testing

Tests interaction between modules.

Examples:

- User module + Database
- Search module + Recommendation engine
- Dashboard + Analytics data

System Testing

Tests the complete application as a single system.

Verification includes:

- Correct navigation
- Recommendation accuracy
- Data consistency
- Dashboard functionality

Performance Testing

Measures:

- Response time
- Search efficiency
- Recommendation speed

Expected result:

- Property search results generated within a few seconds.

Security Testing

Checks:

- Authentication security
- Data protection
- Input validation
- Unauthorized access prevention

User Acceptance Testing (UAT)

Conducted with sample users to ensure:

- Ease of use
- User satisfaction
- Functional correctness

Sample Test Cases

Test Case	Expected Result
User Login	Successful authentication
Property Search	Correct property list displayed
Recommendation Request	Relevant recommendations generated
Analytics Dashboard	Charts displayed correctly
Logout	User session terminated

Table . 4.5 Test Cases

4.6 Maintenance Strategy

Maintenance ensures the system remains functional, secure, and up-to-date after deployment.

Corrective Maintenance

Performed to fix:

- Software bugs
- Database errors
- Interface issues

Adaptive Maintenance

Required when:

- Operating systems change
- Database versions are updated
- New technologies are adopted

Perfective Maintenance

Enhances:

- User interface design
- Recommendation accuracy
- System performance

Preventive Maintenance

Activities include:

- Database backup
- Security updates
- Code optimization

CHAPTER 5: RESULTS AND DISCUSSION

5.1 User Interface Representation

The user interface was designed to be simple, responsive, and easy to navigate. Users can register, log in, search for properties, receive recommendations, and view market analytics without requiring technical knowledge.

Main Features of the User Interface

- Responsive web design
- Easy navigation menu
- Property search filters
- Recommendation section
- Analytics dashboard
- User profile management
- Administrator control panel

5.1: Home Page Interface

The home page serves as the entry point of the application. It contains:

- Navigation bar
- Search bar
- Featured properties
- User login and registration options
- Quick access links

Discussion

The interface provides a clean layout that improves user experience and simplifies property exploration.

5.2 Property Search Module Results

The property search module allows users to search properties using customized filters.

Search Parameters

- Location
- Budget
- Property Type
- Area Size
- Number of Bedrooms
- Amenities

Sample Search Input

Parameter	Value
Location	Bhopal
Budget	₹50,00,000
Property Type	Apartment
Bedrooms	2 BHK

Search Result

Property ID	Location	Type	Price
P101	Bhopal	Apartment	₹48,00,000

Property ID	Location	Type	Price
P105	Bhopal	Apartment	₹49,50,000
P110	Bhopal	Apartment	₹50,00,000

Table no. 5.2.Search Input

5.2.1: Property Search Page

The search page displays filtered properties based on user criteria.

Discussion

The search module efficiently retrieves relevant properties and significantly reduces the effort required for manual searching.

5.3 Recommendation System Results

The recommendation engine analyzes user preferences and generates personalized property suggestions.

Recommendation Factors

- Preferred location
- Budget range
- Property type
- User browsing history
- Property features

Recommendation Output

Rank	Property ID	Matching Score
1	P105	95%
2	P101	92%
3	P110	89%

5.3.1: Recommendation Results Page

Displays recommended properties ranked according to relevance scores.

Discussion

The recommendation engine successfully identifies properties matching user preferences and improves decision-making efficiency. Users receive more relevant suggestions compared to conventional search systems.

5.4 Analytics Dashboard Results

The analytics dashboard provides graphical insights into real estate market conditions.

Dashboard Components

- Property Price Trends
- Location-wise Property Distribution
- Demand Analysis
- Investment Opportunity Analysis
- Property Category Statistics

5.4.1: Analytics Dashboard

The dashboard displays interactive charts and graphs for visual analysis.

Sample Price Trend Analysis

Year	Average Property Price
2021	₹35,00,000
2022	₹38,50,000
2023	₹42,00,000
2024	₹47,00,000
2025	₹52,00,000

3D Property Analysis Graph

The system also provides a 3D graph showing:

- Property Area
- Location Rating
- Property Price

This helps investors identify premium investment opportunities.

Discussion

The analytics dashboard transforms complex property data into meaningful visual information, enabling users to understand market behavior and make informed investment decisions.

Table no. 5.4.Analysis Dashboard

5.5 System Performance Analysis

System performance was evaluated using various testing parameters.

Performance Metrics

Search Response Time

Number of Records	Response Time
1,000	0.5 sec
5,000	1.2 sec
10,000	2.0 sec

Recommendation Generation Time

User Requests	Processing Time
10	0.8 sec
50	1.5 sec
100	2.3 sec

Accuracy Analysis

Recommendation accuracy was evaluated using sample user feedback.

Metric	Value
Recommendation Accuracy	92%
User Satisfaction	90%

Metric	Value
Search Precision	94%

Table no. 5.5. System Performance

5.5.1: Performance Analysis Graph

Graphical representation of:

- Search response time
- Recommendation processing speed
- Accuracy comparison

Discussion

The results indicate that the system performs efficiently even with large datasets. Response times remain acceptable, and recommendation accuracy demonstrates the effectiveness of the implemented algorithms.

5.6 Snapshots with Description

This section presents screenshots of major system modules along with their functionality.

5.6.1: Home Page

Description

- Displays application overview
- Provides navigation menu
- Shows featured properties
- Allows quick property search

5.7: Login and Registration Page

Description

- User registration form
- Secure login interface
- Password validation mechanism

5.8: Property Search Page

Description

- Advanced filtering options
- Search criteria input
- Property listing display

5.9: Property Recommendation Page

Description

- Personalized property suggestions
- Matching score display
- Property details and comparison options

5.10: Analytics Dashboard

Description

- Market trend visualization
- Price prediction graphs
- Demand analysis charts

5.11: User Profile Page

Description

- User account management
- Search history
- Saved properties
- Preference settings

5.12: Admin Dashboard

Description

- Property management
- User management
- Report generation
- System monitoring

CHAPTER 6 :CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Real Estate Analytics and Recommendation System has been successfully designed and developed to provide an intelligent, efficient, and user-friendly platform for property search, analysis, and recommendation. The primary objective of the project was to simplify the real estate decision-making process by integrating data analytics and recommendation techniques into a single web-based application. The developed system enables users to search for properties based on multiple criteria such as location, budget, property type, area, and amenities while receiving personalized recommendations that match their preferences.

The system effectively manages property-related information through a centralized database and provides a structured mechanism for storing, retrieving, and updating property records. The recommendation engine enhances user experience by filtering and ranking properties according to user requirements, thereby reducing the time and effort involved in manually evaluating numerous property listings. In addition, the analytics module offers valuable insights into market trends, price distributions, and investment opportunities through graphical visualizations and statistical analysis.

The implementation of various modules, including user authentication, property management, property search, recommendation generation, and analytics dashboards, demonstrates the practical applicability of modern web technologies and data-driven techniques in the real estate sector. Comprehensive testing was performed to ensure system reliability, security, usability, and performance. The test results indicate that the system functions accurately and efficiently while maintaining acceptable response times and high recommendation relevance.

Furthermore, the project highlights the significance of intelligent decision-support systems in addressing challenges associated with traditional real estate platforms. By providing meaningful recommendations and analytical insights, the system assists buyers, sellers, and investors in making informed decisions and identifying suitable properties more effectively. The developed solution contributes to increased transparency, improved accessibility to real estate information, and enhanced overall user satisfaction.

6.2 Future Scope

Although the proposed Real Estate Analytics and Recommendation System fulfills the current project requirements, several enhancements can be incorporated in future versions to improve functionality, scalability, and recommendation accuracy.

One significant enhancement would be the integration of Artificial Intelligence (AI) and advanced Machine Learning algorithms to provide more accurate and personalized property recommendations. By analyzing user behavior, browsing patterns, and historical interactions, the system can continuously improve recommendation quality and adapt to changing user preferences.

The implementation of property price prediction models using machine learning techniques can further assist investors and buyers in forecasting future property values. Such predictive capabilities would help users evaluate long-term investment opportunities and make more strategic financial decisions.

Future versions may also incorporate Geographic Information System (GIS) technology and interactive maps to provide location-based analysis. Users could visualize nearby facilities such as schools, hospitals, transportation networks, shopping centers, and recreational areas, thereby improving location assessment and property selection.

Another potential enhancement is the development of a mobile application for Android and iOS platforms. Mobile accessibility would allow users to search properties, receive recommendations, and access market analytics from anywhere at any time, improving convenience and user engagement.

The integration of real-time market data from external real estate portals and government property databases could further increase the accuracy and reliability of property information. Automated data synchronization would ensure that users always have access to updated listings and current market conditions.

Additional future improvements may include:

- Virtual property tours using Augmented Reality (AR) and Virtual Reality (VR).
- AI-powered chatbot support for customer assistance.
- Advanced investment risk analysis tools.
- Rental property recommendation and management features.
- Multi-language support for wider accessibility.

- Cloud-based deployment for improved scalability and performance.
- Blockchain-based property verification and secure transaction management.
- Automated report generation for investors and real estate agencies.

By incorporating these advanced technologies and features, the Real Estate Analytics and Recommendation System can evolve into a comprehensive smart real estate platform capable of serving a broader user base and addressing emerging challenges in the rapidly growing real estate industry.

Chapter 7 : REFERENCES / BIBLIOGRAPHY

(IEEE Reference Style for Major Project Report: Real Estate Analytics and Recommendation System)

- [1] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA, USA: Morgan Kaufmann, 2011.
- [2] C. C. Aggarwal, *Recommender Systems: The Textbook*. Cham, Switzerland: Springer International Publishing, 2016.
- [3] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, 3rd ed. Sebastopol, CA, USA: O'Reilly Media, 2022.
- [4] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. New York, NY, USA: McGraw-Hill Education, 2020.
- [5] I. Sommerville, *Software Engineering*, 10th ed. Boston, MA, USA: Pearson Education, 2016.
- [6] A. Silberschatz, H. F. Korth, and S. Sudarshan, *Database System Concepts*, 7th ed. New York, NY, USA: McGraw-Hill Education, 2019.
- [7] R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th ed. Boston, MA, USA: Pearson Education, 2017.
- [8] G. Adomavicius and A. Tuzhilin, "Toward the Next Generation of Recommender Systems: A Survey of the State-of-the-Art and Possible Extensions," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734–749, Jun. 2005.
- [9] F. Ricci, L. Rokach, and B. Shapira, *Recommender Systems Handbook*, 3rd ed. New York, NY, USA: Springer, 2022.
- [10] I. H. Sarker, "Machine Learning: Algorithms, Real-World Applications and Research Directions," *SN Computer Science*, vol. 2, no. 3, pp. 1–21, 2021.

- [11] H. Li and J. Sun, “Real Estate Price Prediction Using Machine Learning Techniques,” *International Journal of Computer Applications*, vol. 176, no. 25, pp. 1–7, 2020.
- [12] S. Kok, P. Monkkonen, and J. M. Quigley, “Land Use Regulations and the Value of Land and Housing: An Intra-Metropolitan Analysis,” *Journal of Urban Economics*, vol. 81, pp. 136–148, 2014.
- [13] Y. Koren, R. Bell, and C. Volinsky, “Matrix Factorization Techniques for Recommender Systems,” *IEEE Computer*, vol. 42, no. 8, pp. 30–37, Aug. 2009.
- [14] M. L. Pazzani and D. Billsus, “Content-Based Recommendation Systems,” in *The Adaptive Web*, Berlin, Germany: Springer, 2007, pp. 325–341.
- [15] X. Su and T. M. Khoshgoftaar, “A Survey of Collaborative Filtering Techniques,” *Advances in Artificial Intelligence*, vol. 2009, pp. 1–19, 2009.
- [16] Python Software Foundation, “Python Documentation,” Available: [Python Official Documentation](#)
- [17] Django Software Foundation, “Django Documentation,” Available: [Django Documentation](#)
- [18] Oracle Corporation, “MySQL Documentation,” Available: [MySQL Documentation](#)
- [19] Scikit-Learn Developers, “Scikit-Learn User Guide,” Available: [Scikit-Learn Documentation](#)
- [20] Plotly Technologies Inc., “Plotly Python Documentation,” Available: [Plotly Documentation](#)

7.1 Books and Research Paper

Books

- [1] Aggarwal, C. C. (2016). *Recommender Systems: The Textbook*. Springer International Publishing.
- [2] Han, J., Kamber, M., & Pei, J. (2011). *Data Mining: Concepts and Techniques* (3rd ed.). Morgan Kaufmann Publishers.
- [3] Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (3rd ed.). O'Reilly Media.
- [4] Pressman, R. S., & Maxim, B. R. (2020). *Software Engineering: A Practitioner's Approach* (9th ed.). McGraw-Hill Education.
- [5] Sommerville, I. (2016). *Software Engineering* (10th ed.). Pearson Education.
- [6] Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts* (7th ed.). McGraw-Hill Education.
- [7] Elmasri, R., & Navathe, S. B. (2017). *Fundamentals of Database Systems* (7th ed.). Pearson Education.

Research Papers

- [8] Adomavicius, G., & Tuzhilin, A. (2005). "Toward the Next Generation of Recommender Systems: A Survey of the State-of-the-Art and Possible Extensions." *IEEE Transactions on Knowledge and Data Engineering*, 17(6), 734–749.
- [9] Ricci, F., Rokach, L., & Shapira, B. (2015). "Recommender Systems Handbook." Springer, New York.
- [10] Sarker, I. H. (2021). "Machine Learning: Algorithms, Real-World Applications and Research Directions." *SN Computer Science*, 2(3), 160.
- [11] Li, H., & Sun, J. (2020). "Real Estate Price Prediction Using Machine Learning Techniques." *International Journal of Computer Applications*, 176(25), 1–7.
- [12] Kok, S., Monkkonen, P., & Quigley, J. M. (2014). "Land Use Regulations and the Value of Land and Housing: An Intra-Metropolitan Analysis." *Journal of Urban Economics*, 81, 136–148.

Chapter 8:APPENDIX- A Source Code Registry

8.1 APPENDIX: Source Code Registry

```
import pandas as pd

import plotly.express as px

import streamlit as st

from utils.analytics import market_kpis, sector_scores
from utils.data_loader import file_status, load_properties
from utils.ui import (
    BUILDING_IMAGES,
    hero,
    inject_global_css,
    metric_card,
    module_card,
    page_config,
    shell_start,
)

page_config("Realty Intelligence")
inject_global_css()
shell_start("Executive Dashboard")


df = load_properties()

hero(
    title="Real Estate",
    accent="Command Center",
    body=(
        "A buyer and investor intelligence platform for pricing, market analysis, "
        "property discovery, and sector-level decision signals."
```

```

    ),
    image_url=BUILDING_IMAGES[0],
    eyebrow="Production-grade analytics for property decisions",)
if df.empty:
    st.error("Could not load the cleaned property dataset. Add the required data file to use the dashboard.")
    st.stop()
kpis = market_kpis(df)
st.markdown('<div class="section-title">Market Pulse</div>', unsafe_allow_html=True)
cols = st.columns(6)
cards = [
    ("Total Listings", kpis["total_properties"], "Cleaned property records"),
    ("Average Price", kpis["avg_price"], "Mean transaction ask"),
    ("Median Price", kpis["median_price"], "Typical market point"),
    ("Avg Price / Sqft", kpis["avg_psf"], "Across available listings"),
    ("Top Sector", kpis["top_sector"], "Highest listing density"),
    ("Dominant Type", kpis["top_type"], "Most common asset class"),
]
for col, (label, value, note) in zip(cols, cards):
    col.markdown(metric_card(label, value, note), unsafe_allow_html=True)

scores = sector_scores(df)
left, right = st.columns([1.15, 0.85])
with left:
    st.markdown('<div class="section-title">Market Snapshot</div>', unsafe_allow_html=True)
    top_snapshot = (
        df.groupby("sector")
        .agg(listings=("sector", "size"), avg_price=("price", "mean"), avg_psf=("price_per_sqft", "mean"))
        .reset_index()
        .sort_values("listings", ascending=False)

```

```

        .head(12)
    )
    fig = px.bar(
        top_snapshot,
        x="sector",
        y="listings",
        color="avg_price",
        color_continuous_scale=["#ffd8c7", "#ff4b2b", "#2b2726"],
        title="Most Active Sectors",
        labels={"sector": "Sector", "listings": "Listings", "avg_price": "Avg Price (Cr)"},
        template="plotly_white",
    )
    fig.update_layout(height=420, margin=dict(l=10, r=10, t=55, b=10), font=dict(color="#2b2726"))
    st.plotly_chart(fig, width="stretch")
with right:
    st.markdown('<div class="section-title">Decision Signals</div>', unsafe_allow_html=True)
    if not scores.empty:
        strongest = scores.iloc[0]
        affordable = scores.sort_values("affordability_score", ascending=False).iloc[0]
        premium = scores.sort_values("avg_price", ascending=False).iloc[0]
        st.markdown(
            metric_card(
                "Best Investment Signal",
                str(strongest["sector"]).title(),
                f"Score {strongest['investment_score']:.0f}/100 with {int(strongest['availability'])} listings",
            ),
            unsafe_allow_html=True,
        )
    st.markdown(

```

```

metric_card(
    "Value Zone",
    str(affordable["sector"]).title(),
    f"Avg psf Rs {affordable['avg_psf']:,.0f}",
),
unsafe_allow_html=True,
)
st.markdown(
    metric_card(
        "Premium Zone",
        str(premium["sector"]).title(),
        f"Avg price Rs {premium['avg_price']:.2f} Cr",
    ),
    unsafe_allow_html=True,
)

st.markdown('<div class="section-title">Product Modules</div>', unsafe_allow_html=True)
m1, m2, m3, m4 = st.columns(4)
m1.markdown(
    module_card(
        "Price Prediction",
        "Estimate fair market value using the existing ML pipeline and production-style validation.",
        "ML Valuation",
    ),
    unsafe_allow_html=True,
)
m2.markdown(
    module_card(
        "Market Analysis",

```



```

        "Filter inventory and compare sectors, property types, price bands, and luxury signals.",
        "Analytics",
    ),
    unsafe_allow_html=True,
)
m3.markdown(
    module_card(
        "Recommendations",
        "Discover similar societies and location-radius matches with image-led property cards.",
        "Discovery",
    ),
    unsafe_allow_html=True,
)
m4.markdown(
    module_card(
        "Advanced Insights",
        "Use sector scoring, persona recommendations, outlier detection, and society rankings.",
        "Strategy",
    ),
    unsafe_allow_html=True,
)

st.markdown('<div class="section-title">Deployment Readiness</div>', unsafe_allow_html=True)
status = file_status()
status_df = pd.DataFrame(
    [{"Artifact": name.replace("_", " ").title(), "Available": "Ready" if ok else "Missing"} for name, ok in
status.items()]
)
st.dataframe(status_df, width="stretch", hide_index=True)

```

